IASYN AI

AI-Powered Rehabilitation Monitoring System

Intelligent Movement. Smarter Recovery.

TEAM AROGYA

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PROBLEM STATEMENT

Challenges in Home Rehabilitation Monitoring

Hard to monitor rehabilitation exercises at home

Limited continuous physiotherapist supervision

Difficult to track exercise performance and recovery

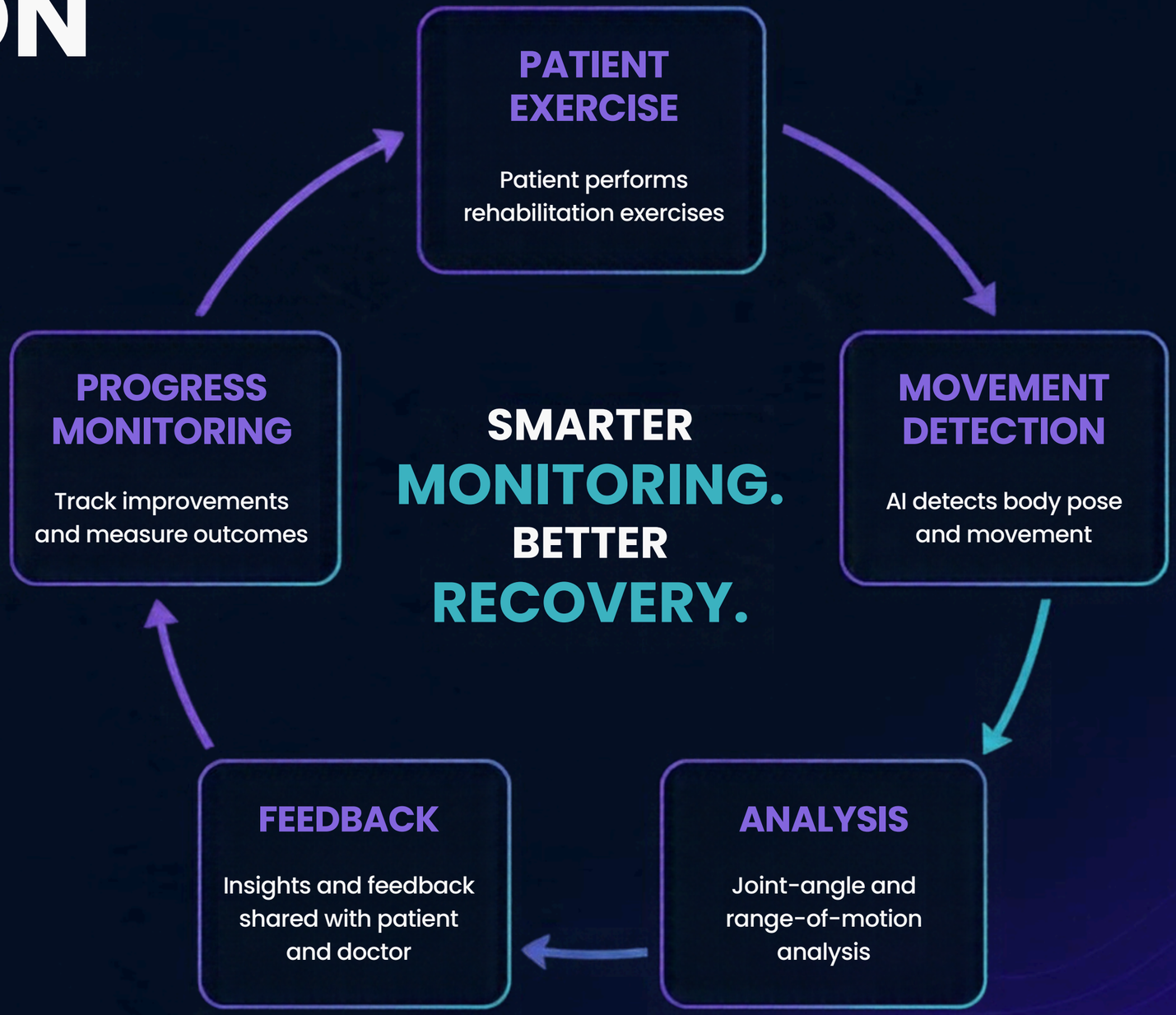
GOAL:

Make rehabilitation monitoring smarter and more accessible

OUR SOLUTION

AI-Powered Rehabilitation Monitoring Features

- Camera-based AI pose detection for accurate movement tracking
- Joint-angle and range-of-motion analysis for precise assessment
- Exercise repetition and movement tracking with progress reports
- Patient progress reports and doctor feedback integration
- Leg position monitoring via sensor-based audio feedback



SYSTEM ARCHITECTURE

Implemented and Planned Components

SENSOR INPUT

ESP32 + IR Sensor +
Ultrasonic Sensor + Buzzer
(Prototype)

AI PROCESSING

Joint-angle calculations
Range-of-motion estimation
Movement analysis

SOFTWARE STACK

- React
- TypeScript
- Vite
- Tailwind CSS
- FastAPI
- Database

OUTPUT & FEEDBACK

Patient dashboard
Movement reports
Clinician feedback
Local audio feedback

PLANNED ENHANCEMENTS

Expanded AI analysis
Additional sensors
Advanced insights
Cloud integration

SOFTWARE IMPLEMENTATION

Smart application features for end-to-end rehabilitation monitoring

Patient dashboard for monitoring and assessments

Knee rehabilitation exercise guide with detailed instructions

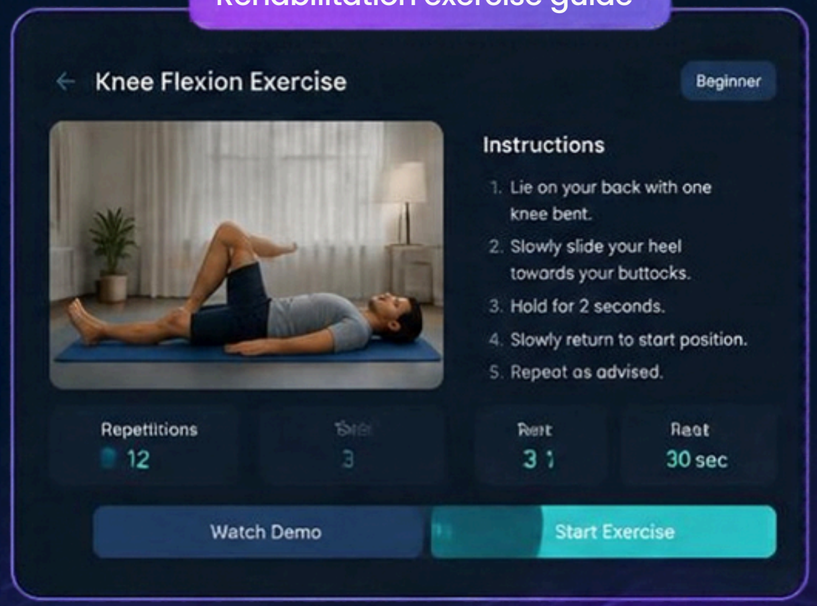
Live movement assessment for real-time feedback

Patient movement reports summarizing progress

Doctor and administrator portals for clinical use

Patient dashboard

Rehabilitation exercise guide



Patient movement reports

ESP32-Based Leg Rehabilitation Feedback System

HARDWARE COMPONENTS

ESP32

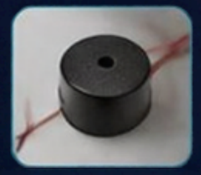
Controls sensors and buzzer

IR SENSOR

Detects nearby objects or movement depending on placement

ULTRASONIC SENSOR

Measures distance to monitor exercise movement



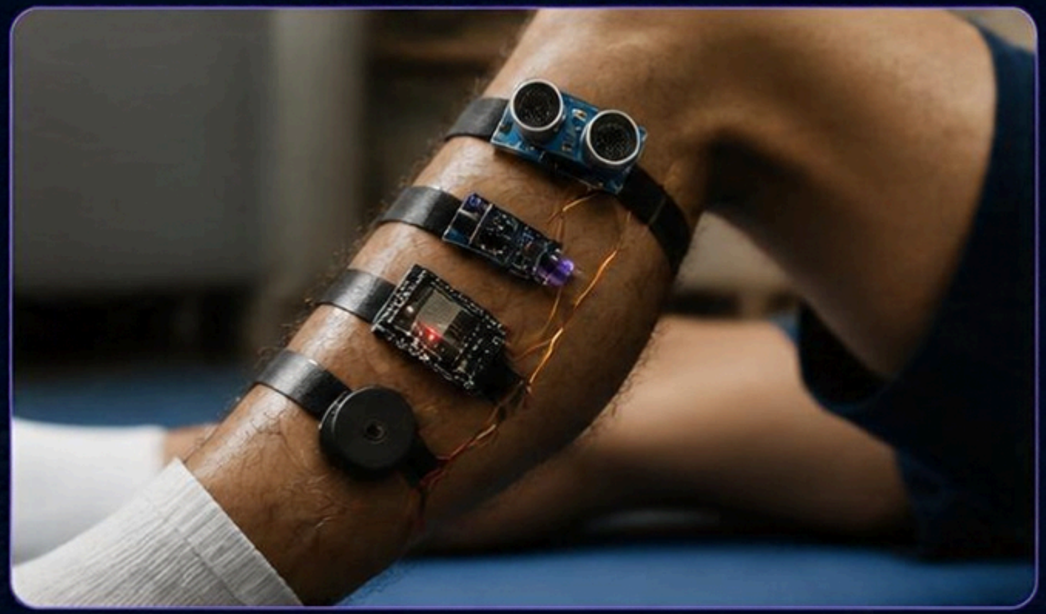
BUZZER

Provides audio alerts on feedback conditions

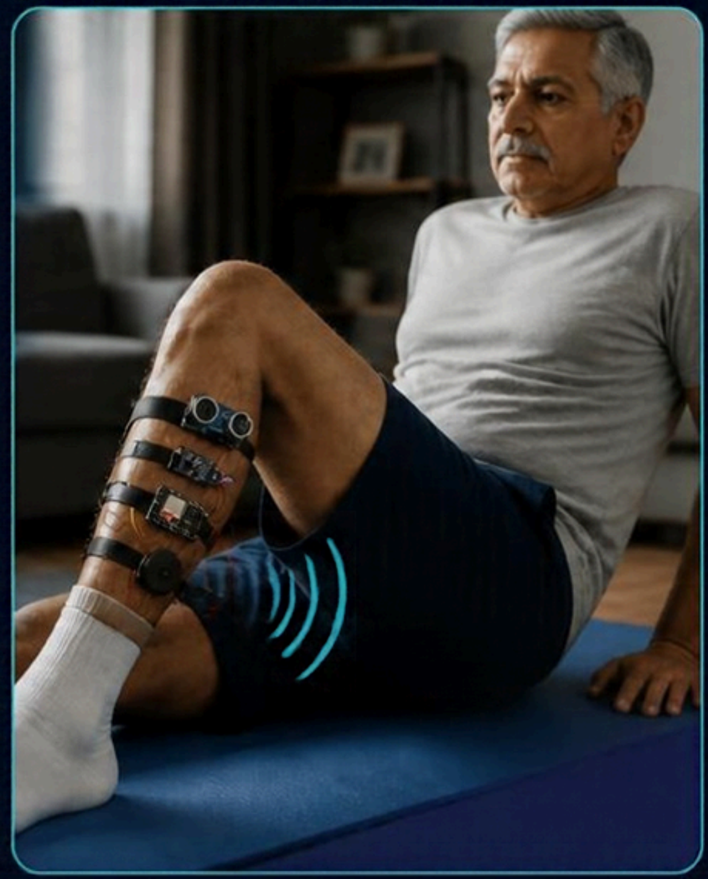
WORKING FLOW

IR Sensor + Ultrasonic Sensor → ESP32 → Condition Checking → Buzzer Alert

ACTUAL PROTOTYPE SETUP



REHABILITATION EXERCISE



IMPORTANT NOTE: Sensors assist in monitoring movement patterns and proximity.
They do not directly measure joint angles.

RESULTS AND FUTURE SCOPE

CURRENT ACHIEVEMENTS

- Rehabilitation dashboard and exercise guide
- Patient assessment history and reports
- Clinician feedback interface
- ESP32-based sensor and buzzer prototype

NEXT STEPS

- Support additional rehabilitation exercises
- Improve movement analysis algorithms
- Expand exercise recognition capabilities
- Validation and collaboration with physiotherapists

MAKING REHABILITATION
SMARTER, MEASURABLE, AND ACCESSIBLE.